

Sharpness of the $(1 + t)$ factor in nested ribbon commutators, for all k

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Abstract

Let $R_e(t)$ be the height-graded e -ribbon operator on the Fock space $\Lambda = \mathbb{Q}(t) \otimes \Lambda_{\mathbb{Z}}$, and for $e_1, \dots, e_k \geq 2$ let $\mathcal{C}_k = [R_{e_1}, [R_{e_2}, [\dots, [R_{e_{k-1}}, R_{e_k}] \dots]]]$. It is known (Q81, Theorem 1) that $1 + t$ divides every matrix entry of $\mathcal{C}_k$ for every k , and (Q81, Theorem 2) that the divisibility is sharp at $k = 3$ when the three sizes are distinct and the largest is not outermost. We prove sharpness for *all* k , and we remove both hypotheses.

The mechanism is a closed form. Writing $E = \sum_i e_i$, we show that for *every* j with $0 \leq j \leq E - 1$ the single matrix entry of $\mathcal{C}_k$ at the hook $\mu = (E - j, 1^j)$, taken from the vacuum, equals

$$\langle (E - j, 1^j) \mid \mathcal{C}_k \mid \emptyset \rangle = \operatorname{sgn}(e_k - e_{k-1}) (1 + t) \sum_{\substack{T \subseteq \{1, \dots, k-2\} \\ \Sigma_T + e_{\min} \leq j < \Sigma_T + e_{\max}}} t^{j-1-|T|},$$

where $\Sigma_T = \sum_{i \in T} e_i$ and $e_{\min}, e_{\max}$ are the smaller and larger of e_{k-1}, e_k . Dividing by $1 + t$ and evaluating at $t = -1$ collapses this to a single polynomial coefficient:

$$\left\langle (E - j, 1^j) \mid \frac{\mathcal{C}_k}{1+t} \mid \emptyset \right\rangle_{t=-1} = (-1)^{j-1} [x^j] \Phi_{\vec{e}}(x), \quad \Phi_{\vec{e}}(x) := \frac{x^{e_{k-1}} - x^{e_k}}{1 - x} \prod_{i=1}^{k-2} (1 - x^{e_i}).$$

Since $\mathbb{Z}[x]$ is an integral domain, $\Phi_{\vec{e}} \neq 0$ as soon as $e_{k-1} \neq e_k$; some coefficient is nonzero, and that j furnishes a witness. Conversely $e_{k-1} = e_k$ forces $\mathcal{C}_k = 0$. So the $(1 + t)$ -adic valuation of the gcd of the entries of $\mathcal{C}_k$ is exactly 1 if and only if $e_{k-1} \neq e_k$ — with no distinctness assumption on the remaining sizes and no condition on where the largest size sits. All closed forms are verified entrywise against an independent bead-model engine (149 + 97 entries, 0 mismatches) with a planted-error control.

We also record what the previous session's evidence was and was not measuring: the witness family of Q81 is the specialisation $j = f_{k-1}$ of the above, and for $k \geq 4$ that specialisation *degenerates* — the entry can vanish, or acquire a second factor of $1 + t$. The $k \leq 6$ verification that supported the conjecture was run on sorted tuples, where $e_{k-1} = f_{k-1} = j$ forces the sum to its single term $T = \emptyset$; the test was constant in the direction it appeared to probe.

1 Setting, and what is inherited

Fix an indeterminate t . For a partition λ let $|\lambda\rangle = s_{\lambda}$, and let $R_e(t)$ be the operator

$$R_e(t) |\lambda\rangle = \sum_{\mu} t^{h(\mu/\lambda)} |\mu\rangle,$$

the sum over $\mu \supseteq \lambda$ such that μ/λ is a connected border strip (ribbon) of size e , with h its height (rows minus one). In the Maya (bead) model, $\lambda \leftrightarrow B(\lambda) = \{\lambda_i - i\}$, this is the sum over legal bead hops $b \mapsto b + e$ weighted by $t^{\#(B \cap (b, b+e))}$. At $t = -1$ it is the Murnaghan–Nakayama operator, i.e. multiplication by p_e .

Throughout, $e_1, \dots, e_k \geq 2$, $E := \sum_i e_i$, and

$$\mathcal{C}_k = \mathcal{C}_k(e_1, \dots, e_k) := [R_{e_1}, [R_{e_2}, [\dots, [R_{e_{k-1}}, R_{e_k}] \dots]]].$$

We use four results from the previous paper (2026-09-04-Q81-nested-bracket.tex, all graded proved there).

Theorem 1 (Q81, Thm. thm:div). *$1+t$ divides every matrix entry of $\mathcal{C}_k$, for every $k \geq 2$ and all $e_i \geq 2$, with no distinctness hypothesis.*

Theorem 2 (Q81, Thm. thm:wit). *For $k = 3$ and e_1, e_2, e_3 pairwise distinct with sorted values $f_1 < f_2 < f_3$ and $j = f_2$, the entry $\langle (E-j, 1^j) | \mathcal{C}_3 | \emptyset \rangle$ equals $t^{j-1}(1+t)$, $-t^{j-1}(1+t)$, or 0 according as the largest size sits in slot 3, 2, or 1.*

The next three are Lemmas `lem:chain`, `lem:legal`, `lem:height` and `lem:sign` of Q81. We restate them with the parameter j left free; the proofs given there use only $-1-j < 0 \leq E-1-j$, i.e. $0 \leq j \leq E-1$, and never the choice $j = f_{k-1}$. We take them in this form.

Lemma 3 (chain, legality, height). *Let $\lambda = \emptyset$, so $B = \mathbb{Z}_{<0}$; fix j with $0 \leq j \leq E-1$, put $b = -1-j$ and $\mu = (E-j, 1^j)$, so that $B(\mu) = B \setminus \{b\} \cup \{b+E\}$. Then:*

- (a) *Every sequence of k legal bead hops taking B to $B(\mu)$, one performed by each letter of a word $R_{e_{w_1}} \cdots R_{e_{w_k}}$, is a spatial chain: there is a permutation $\rho \in S_k$ and $b = x_0 < x_1 < \cdots < x_k = b+E$ with $x_i - x_{i-1} = e_{\rho(i)}$, the i th spatial hop moving a bead from x_{i-1} to x_i .*
- (b) *Put $r = r(\rho) := \min\{i : e_{\rho(1)} + \cdots + e_{\rho(i)} > j\}$. A time order of the k spatial hops is legal if and only if it is a linear extension of the poset $P^{(r)}$ on $\{1, \dots, k\}$ generated by $r \prec r-1 \prec \cdots \prec 1$ and $r \prec r+1 \prec \cdots \prec k$.*
- (c) *Every legal time order of the chain ρ contributes weight $t^{j-r(\rho)+1}$; in particular $r(\rho) \leq j+1$.*

Lemma 4 (Q81, `lem:sign`). *Expanding $[A_1, [A_2, [\dots, [A_{k-1}, A_k] \dots]]]$ into words, the words occurring are exactly those whose time-ordered index sequence $w = (w_1, \dots, w_k)$ (rightmost factor first) is unimodal with peak k : $w_1 < \cdots < w_q = k > w_{q+1} > \cdots > w_k$. Each occurs once, with sign $\varepsilon(w) = (-1)^{q-1}$.*

Assembling Lemmas 3 and 4:

$$\langle (E-j, 1^j) | \mathcal{C}_k | \emptyset \rangle = \sum_{\rho \in S_k} \Sigma(\rho, r(\rho)) t^{j-r(\rho)+1}, \quad \Sigma(\rho, r) := \sum_{\tau \in \mathcal{L}(P^{(r)})} \varepsilon(\rho \circ \tau), \quad (1)$$

where $\mathcal{L}(P^{(r)})$ is the set of linear extensions of $P^{(r)}$, written as words, and $\rho \circ \tau$ denotes the word $(\rho(\tau_1), \dots, \rho(\tau_k))$. Note that $\mathcal{L}(P^{(r)})$ consists of the words beginning with r and continuing with an arbitrary shuffle of the decreasing run $(r-1, r-2, \dots, 1)$ and the increasing run $(r+1, r+2, \dots, k)$.

Everything in this paper is the evaluation of (1).

2 The sign sum is a single term

Write $U \subseteq S_k$ for the set of permutations which, read as words, are unimodal with peak value k ; $|U| = 2^{k-1}$, and $\rho \in U$ is determined by the set $T = \{\rho(1), \dots, \rho(q-1)\} \subseteq \{1, \dots, k-1\}$ of values preceding k , via

$$\rho_T = (T \text{ increasing, } k, (\{1, \dots, k-1\} \setminus T) \text{ decreasing}), \quad \varepsilon(\rho_T) = (-1)^{|T|}. \quad (2)$$

Proposition 5 (the sign sum). *For every $\rho \in S_k$ and every $1 \leq r \leq k$,*

$$\Sigma(\rho, r) = (-1)^{r-1} \varepsilon(\rho).$$

In particular $\Sigma(\rho, r) = 0$ unless $\rho \in U$, and then $\Sigma(\rho_T, r) = (-1)^{r-1}(-1)^{|T|}$.

Proof. Write $\rho = (c_1, \dots, c_k)$ as a word and let $p := \rho^{-1}(k)$ be the position of the maximal value. A linear extension τ starts at r and thereafter interleaves the *down* run $r-1, r-2, \dots, 1$ with the *up* run $r+1, r+2, \dots, k$; thus $w := \rho \circ \tau$ is the word c_r followed by a shuffle of $(c_{r-1}, \dots, c_1)$ and $(c_{r+1}, \dots, c_k)$. We must decide when w is unimodal with peak k , and sum $(-1)^{q-1}$ over those τ .

Case $p = r$. Then $w_1 = k$, so $q = 1$ and $\varepsilon(w) = +1$ provided w is strictly decreasing after position 1. A shuffle of two sequences is strictly decreasing iff both are strictly decreasing, and then the shuffle realising it is unique. So there is a contributing τ iff $c_{r-1} > c_{r-2} > \dots > c_1$ fails to hold — rather, iff $(c_{r-1}, \dots, c_1)$ and $(c_{r+1}, \dots, c_k)$ are both strictly decreasing, i.e. iff $c_1 < c_2 < \dots < c_{r-1}$ and $c_{r+1} > c_{r+2} > \dots > c_k$; and then it is unique and contributes $+1$. These conditions, together with $c_r = k$, say exactly that $\rho \in U$ with $q(\rho) = r$, i.e. $\varepsilon(\rho) = (-1)^{r-1}$. Hence $\Sigma(\rho, r) = (-1)^{r-1} \varepsilon(\rho)$ in both the contributing and the non-contributing case.

Case $p < r$. The value $k = c_p$ is consumed by τ only after the down-run has passed $r-1, \dots, p+1$. Let $s \in \{0, 1, \dots, k-r\}$ be the number of up-run letters placed before k . Then the letters of w preceding k are c_r , then a shuffle of $(c_{r-1}, \dots, c_{p+1})$ and $(c_{r+1}, \dots, c_{r+s})$, and the letters following k are a shuffle of $(c_{p-1}, \dots, c_1)$ and $(c_{r+s+1}, \dots, c_k)$. Requiring w unimodal with peak k therefore forces, and is forced by:

- (C1) $c_r < c_{r-1} < \dots < c_{p+1}$, i.e. ρ strictly decreasing on $[p, r]$ (automatic at p , since $c_p = k$ is maximal);
- (C2) $c_{r+1} < c_{r+2} < \dots < c_{r+s}$, and $c_r < c_{r+1}$ if $s \geq 1$;
- (C3) $c_{p-1} > c_{p-2} > \dots > c_1$, i.e. ρ strictly increasing on $[1, p]$;
- (C4) $c_{r+s+1} > c_{r+s+2} > \dots > c_k$;

and, given s , the shuffle is then unique (a merge of two increasing, resp. two decreasing, sequences of distinct values is unique). The position of k is $q = 1 + (r - p - 1) + s + 1 = r - p + s + 1$, so $\varepsilon(w) = (-1)^{r-p+s}$. Hence, if (C1) and (C3) hold,

$$\Sigma(\rho, r) = (-1)^{r-p} \sum_{s \in A} (-1)^s, \quad A := \{s : (\text{C2}) \text{ and } (\text{C4}) \text{ hold}\},$$

and $\Sigma(\rho, r) = 0$ otherwise. If (C1) or (C3) fails then ρ is not unimodal, so $\varepsilon(\rho) = 0$ and the claim holds trivially. Assume (C1) and (C3). Then ρ increases on $[1, p]$ and decreases on $[p, r]$, so $\rho \in U$ if and only if in addition $c_r > c_{r+1} > \dots > c_k$. Put $n := k - r$ and $u := (c_{r+1}, \dots, c_k)$.

If $c_r > c_{r+1}$ (or $n = 0$), then (C2) forbids $s \geq 1$, so $A \subseteq \{0\}$ and $0 \in A$ iff u is strictly decreasing. Thus $A = \{0\}$ and $\Sigma(\rho, r) = (-1)^{r-p}$ exactly when $\rho \in U$, in which case $q(\rho) = p$ and $\varepsilon(\rho) = (-1)^{p-1}$, giving $\Sigma(\rho, r) = (-1)^{r-p} = (-1)^{r-1}(-1)^{p-1} = (-1)^{r-1}\varepsilon(\rho)$; and $A = \emptyset$, $\Sigma = 0 = \varepsilon(\rho)$ otherwise.

If $c_r < c_{r+1}$ (so $n \geq 1$ and $\rho \notin U$, since ρ has an ascent at r after its peak), the constraint in (C2) is vacuous and $A = \{s : u_1 < \dots < u_s \text{ and } u_{s+1} > \dots > u_n\}$. Let α be the length of the initial strictly increasing run of u and β the least index with $u_\beta > u_{\beta+1} > \dots > u_n$ (so $1 \leq \beta \leq n$ and $1 \leq \alpha \leq n$). Then $A = \{s : \beta - 1 \leq s \leq \alpha\}$. We claim $\beta = \alpha$ whenever $A \neq \emptyset$. Indeed $\beta \leq \alpha$ is impossible, since it would give both $u_\beta > u_{\beta+1}$ and $u_\beta < u_{\beta+1}$ (using $\beta + 1 \leq \alpha$; and $\beta = \alpha$ is the excluded boundary), so $\beta \geq \alpha$; while $\beta = \alpha + 1$ would require $u_\alpha < u_{\alpha+1}$ (else $\beta \leq \alpha$), contradicting maximality of the initial increasing run, which forces $u_\alpha > u_{\alpha+1}$ whenever $\alpha < n$ — and if $\alpha = n$ then $\beta \leq n = \alpha$. Hence $A \neq \emptyset$ implies $\beta = \alpha$ and $A = \{\alpha - 1, \alpha\}$, so $\sum_{s \in A} (-1)^s = 0 = \varepsilon(\rho)$, as required.

Case $p > r$. Set $\hat{\tau}_l := k + 1 - \tau_l$ and $\hat{\rho}(i) := \rho(k + 1 - i)$, and $r' := k + 1 - r$. Then $\tau \mapsto \hat{\tau}$ is a bijection $\mathcal{L}(P^{(r)}) \rightarrow \mathcal{L}(P^{(r')})$ (it exchanges the down and up runs), and $\hat{\rho} \circ \hat{\tau} = \rho \circ \tau$, so $\Sigma(\hat{\rho}, r') = \Sigma(\rho, r)$. Moreover $\hat{\rho}^{-1}(k) = k + 1 - p < k + 1 - r = r'$, so the pair $(\hat{\rho}, r')$ falls under the case already proved. Finally $\hat{\rho}$ is the reversal of ρ , and reversal maps a unimodal word with peak at q to a unimodal word with peak at $k + 1 - q$, so $\varepsilon(\hat{\rho}) = (-1)^{k-q} = (-1)^{k-1}\varepsilon(\rho)$. Therefore

$$\Sigma(\rho, r) = \Sigma(\hat{\rho}, r') = (-1)^{r'-1}\varepsilon(\hat{\rho}) = (-1)^{k-r}(-1)^{k-1}\varepsilon(\rho) = (-1)^{r-1}\varepsilon(\rho). \quad \square$$

3 The hook entry in closed form

Theorem 6 (closed form for the hook entries). *Let $k \geq 2$ and $e_1, \dots, e_k \geq 2$ (no distinctness assumed). Put $E = \sum_i e_i$, $e_{\min} = \min(e_{k-1}, e_k)$, $e_{\max} = \max(e_{k-1}, e_k)$ and, for $T \subseteq \{1, \dots, k-2\}$, $\Sigma_T = \sum_{i \in T} e_i$. Then for every j with $0 \leq j \leq E - 1$,*

$$\langle (E - j, 1^j) \mid \mathcal{C}_k \mid \emptyset \rangle = \text{sgn}(e_k - e_{k-1}) (1 + t) \sum_{\substack{T \subseteq \{1, \dots, k-2\} \\ \Sigma_T + e_{\min} \leq j < \Sigma_T + e_{\max}}} t^{j-1-|T|}. \quad (3)$$

(For $e_{k-1} = e_k$ both sides are 0.)

Proof. By Proposition 5 only $\rho \in U$ contribute to (1), so by (2)

$$\langle (E - j, 1^j) \mid \mathcal{C}_k \mid \emptyset \rangle = \sum_{T \subseteq \{1, \dots, k-1\}} (-1)^{|T|} (-1)^{r_T-1} t^{j+1-r_T}, \quad r_T := r(\rho_T). \quad (4)$$

Note r_T is well defined because the total $E > j$.

Pair the subsets of $\{1, \dots, k-1\}$ by toggling the element $k-1$: each T lies in exactly one pair $\{T_0, T_1\}$ with $T_0 \subseteq \{1, \dots, k-2\}$ and $T_1 = T_0 \cup \{k-1\}$. Put $a := |T_1| = |T_0| + 1$ and $s := \Sigma_{T_0}$. Because $k-1$ is the largest element of $\{1, \dots, k-1\}$, it occupies the last slot of the increasing block of ρ_{T_1} and the first slot of the decreasing block of ρ_{T_0} ; explicitly

$$\rho_{T_1} = (T_0 \uparrow, k-1, k, D \downarrow), \quad \rho_{T_0} = (T_0 \uparrow, k, k-1, D \downarrow), \quad D := \{1, \dots, k-2\} \setminus T_0,$$

so ρ_{T_0} and ρ_{T_1} differ exactly by transposing the entries in positions a and $a+1$. Consequently their sequences of partial sums of sizes agree in every position except position a , where they are $s + e_{k-1}$ (for T_1) and $s + e_k$ (for T_0).

If $s > j$, the threshold is already crossed before position a , so $r_{T_0} = r_{T_1}$; the two terms of (4) then differ only by the factor $(-1)^{|T|}$ and cancel. Suppose $s \leq j$. If $s + e_{k-1}$ and $s + e_k$ are both $> j$ then $r_{T_0} = r_{T_1} = a$ and again the terms cancel; if both are $\leq j$ then the partial sums agree from position $a + 1$ onwards (both equal $s + e_{k-1} + e_k$ there) and beyond, so $r_{T_0} = r_{T_1}$ and the terms cancel.

There remains the case in which exactly one of $s + e_{k-1}$, $s + e_k$ exceeds j , i.e.

$$s + e_{\min} \leq j < s + e_{\max}. \quad (5)$$

If $e_k > e_{k-1}$ then $r_{T_0} = a$ and $r_{T_1} = a + 1$, and the two terms of (4) are $(-1)^{a-1}(-1)^{a-1}t^{j+1-a} = t^{j+1-a}$ and $(-1)^a(-1)^at^{j-a} = t^{j-a}$, with sum $t^{j-a}(1+t)$. If $e_{k-1} > e_k$ then $r_{T_1} = a$ and $r_{T_0} = a + 1$, and the terms are $(-1)^a(-1)^{a-1}t^{j+1-a} = -t^{j+1-a}$ and $(-1)^{a-1}(-1)^at^{j-a} = -t^{j-a}$, with sum $-t^{j-a}(1+t)$. In both cases the pair contributes $\text{sgn}(e_k - e_{k-1})t^{j-a}(1+t)$, and $j - a = j - 1 - |T_0|$. Summing over the surviving pairs, indexed by $T_0 =: T \subseteq \{1, \dots, k-2\}$ subject to (5), gives (3). If $e_{k-1} = e_k$ then (5) is empty and the right side vanishes, as does $\mathcal{C}_k$ itself. $\square$

Remark 7. Theorem 6 re-proves Theorem 1 for this family of entries, without using the decomposition $R_e = M_{f_e} - (1+t)E_e$: the factor $1+t$ appears because the toggling involution leaves exactly the pairs on which r jumps by one, and a jump of r by one multiplies the weight t^{j+1-r} by t^{-1} .

4 Sharpness for all k

Theorem 8 (main). *Let $k \geq 2$ and $e_1, \dots, e_k \geq 2$, and set*

$$\Phi_{\vec{e}}(x) := \frac{x^{e_{k-1}} - x^{e_k}}{1 - x} \prod_{i=1}^{k-2} (1 - x^{e_i}) \in \mathbb{Z}[x].$$

Then for every $0 \leq j \leq E - 1$,

$$\left\langle (E - j, 1^j) \left| \frac{\mathcal{C}_k}{1 + t} \right| \emptyset \right\rangle_{t=-1} = (-1)^{j-1} [x^j] \Phi_{\vec{e}}(x). \quad (6)$$

Consequently the following are equivalent:

- (i) $e_{k-1} \neq e_k$;
- (ii) $\mathcal{C}_k \neq 0$;
- (iii) *the $(1+t)$ -adic valuation of gcd of the matrix entries of $\mathcal{C}_k$ is exactly 1, i.e. $1+t$ divides every entry and $(1+t)^2$ does not divide all of them.*

No distinctness of $e_1, \dots, e_{k-2}$ is required, and no condition is placed on the position of the largest size.

Proof. By Theorem 6, dividing (3) by $1 + t$ and setting $t = -1$ gives

$$\text{sgn}(e_k - e_{k-1}) \sum_{T: \Sigma_T + e_{\min} \leq j < \Sigma_T + e_{\max}} (-1)^{j-1-|T|} = (-1)^{j-1} \text{sgn}(e_k - e_{k-1}) \sum_{T: j - e_{\max} < \Sigma_T \leq j - e_{\min}} (-1)^{|T|}.$$

Now $\sum_{T \subseteq \{1, \dots, k-2\}} (-1)^{|T|} x^{\Sigma T} = \prod_{i \leq k-2} (1 - x^{e_i})$, and summing the coefficients of that product over the window $j - e_{\max} < d \leq j - e_{\min}$ is the same as extracting $[x^j]$ after multiplying by $x^{e_{\min}} + x^{e_{\min}+1} + \dots + x^{e_{\max}-1} = \frac{x^{e_{\min}} - x^{e_{\max}}}{1 - x}$. Since $\text{sgn}(e_k - e_{k-1}) \cdot \frac{x^{e_{\min}} - x^{e_{\max}}}{1 - x} = \frac{x^{e_{k-1}} - x^{e_k}}{1 - x}$ in both orderings, this is $(-1)^{j-1} [x^j] \Phi_{\vec{e}}$, proving (6).

For the equivalences: if $e_{k-1} = e_k$ then $[R_{e_{k-1}}, R_{e_k}] = 0$, so $\mathcal{C}_k = 0$ and (ii),(iii) both fail. If $e_{k-1} \neq e_k$ then $x^{e_{k-1}} - x^{e_k} \neq 0$ and each $1 - x^{e_i} \neq 0$ in the integral domain $\mathbb{Z}[x]$, so $\Phi_{\vec{e}} \neq 0$: some coefficient $[x^{j_0}] \Phi_{\vec{e}}$ is nonzero. Every nonzero coefficient of $\Phi_{\vec{e}}$ lies in degrees between $e_{\min} \geq 2$ and $\deg \Phi_{\vec{e}} = e_{\max} - 1 + \sum_{i \leq k-2} e_i = E - e_{\min} - 1$, so $0 \leq j_0 \leq E - 1$ and $(E - j_0, 1^{j_0})$ is a genuine partition of E . By (6) the entry of $\mathcal{C}_k$ at that hook is divisible by $1+t$ but not by $(1+t)^2$; combined with Theorem 1 this gives (iii), and a fortiori (ii). $\square$

Corollary 9 (Q83 as posed). *For every $k \geq 2$ and every pairwise distinct $e_1, \dots, e_k \geq 2$, the $(1+t)$ -adic valuation of $\gcd(\mathcal{C}_k)$ is exactly 1. The hypothesis $\max_i \{e_i\} \neq e_1$ of the $k = 3$ result is unnecessary.*

Corollary 10 (an explicit uniform witness). *Suppose $\{e_{k-1}, e_k\}$ contains the largest size, and $\min(e_{k-1}, e_k) + \min_{i \leq k-2} e_i > f_{k-1}$, where $f_1 \leq \dots \leq f_k$ is the sorted size list. Then, with $j = f_{k-1}$, the window (5) admits only $T = \emptyset$ and*

$$\langle (E - f_{k-1}, 1^{f_{k-1}}) \mid \mathcal{C}_k \mid \emptyset \rangle = \text{sgn}(e_k - e_{k-1}) t^{f_{k-1}-1} (1+t).$$

In particular this holds whenever the two largest sizes occupy the two innermost slots.

Proof. $e_{\max} = f_k > f_{k-1} = j$ gives $\Sigma_{\emptyset} = 0 > j - e_{\max}$, and $e_{\min} \leq j$ gives $0 \leq j - e_{\min}$, so $\emptyset$ lies in the window; the second hypothesis says $\Sigma_T \geq \min_{i \leq k-2} e_i > j - e_{\min}$ for every nonempty T , excluding all others. When $\{e_{k-1}, e_k\} = \{f_{k-1}, f_k\}$ one has $e_{\min} = f_{k-1} = j$ and the second hypothesis reads $\min_{i \leq k-2} e_i > 0$. $\square$

5 What the previous evidence was measuring

Theorem 2 is exactly the specialisation $j = f_{k-1}$ of Theorem 6 at $k = 3$: there $\{1, \dots, k-2\} = \{1\}$, and one checks directly that the window (5) contains $\emptyset$ iff the largest size is in slot 2 or 3, and contains $\{1\}$ never. For $k \geq 4$ the same specialisation *degenerates*.

Example 11 (the hook witness fails at $k = 4$). Take $(e_1, e_2, e_3, e_4) = (2, 5, 3, 6)$, so $f = (2, 3, 5, 6)$, $j = f_3 = 5$, $E = 16$. The window (5) is $\{T \subseteq \{1, 2\} : \Sigma_T + 3 \leq 5 < \Sigma_T + 6\} = \{\emptyset, \{1\}\}$, so

$$\langle (11, 1^5) \mid \mathcal{C}_4 \mid \emptyset \rangle = (1+t)(t^4 + t^3) = t^3(1+t)^2,$$

which is divisible by $(1+t)^2$ and therefore carries no information about sharpness — even though the largest size is innermost. Similarly, at $k = 4$ the entry vanishes identically for 74 of the 90 tuples with the largest size in slot 2 (e.g. $(2, 5, 3, 4)$). Sharpness for $(2, 5, 3, 6)$ is instead witnessed at $j = 8$, where $[x^8] \Phi_{\vec{e}} \neq 0$.

Example 12 (the excluded case of Theorem 2 is not an obstruction). For $(e_1, e_2, e_3) = (4, 2, 3)$ the largest size is outermost and the $j = f_2 = 3$ entry is 0, which is why Q81 excluded it. But $\Phi_{\vec{e}}(x) = \frac{x^2 - x^3}{1-x} (1 - x^4) = x^2(1 - x^4) = x^2 - x^6$, so $j = 2$ gives $\langle (7, 1, 1) \mid \mathcal{C}_3 \mid \emptyset \rangle = t(1+t)$ and $j = 6$ gives $\langle (3, 1^6) \mid \mathcal{C}_3 \mid \emptyset \rangle = t^4(1+t)$. Sharpness holds.

Remark 13 (why the $k \leq 6$ verification was flat). Q81 reports the entry to be $t^{j-1}(1+t)$ for every 4- and 5-subset of $\{2, \dots, 6\}$ tested with the largest size innermost. Those tests used the sizes in increasing order, so $e_{k-1} = f_{k-1} = j$ and $e_{\min} = j$; the window (5) then reads $\Sigma_T \leq 0 < \Sigma_T + f_k$, forcing $T = \emptyset$ for *every* such tuple. The observed constancy of the exponent was a property of the arrangement tested, not evidence about k : the test was constant in the direction it appeared to probe. Example 11 is what varying the arrangement produces.

6 An independent reading of the level-2 operator

The session brief proposed reducing $\mathcal{C}_k$, via the Q81 reduction formula, to the differential order of the single operator $N_e := E_e(-1)$ — the operator adding a size- e generalized border strip with exactly two components, weighted $(-1)^{\text{height}}$. That route is not needed for Theorem 8, but the following identity, obtained on the way, identifies N_e and is recorded because it answers the brief's Phase 0 question outright.

Proposition 14. *Let M_g denote multiplication by $g \in \Lambda$, let $\alpha_{-a} = M_{p_a}$, and let $\mathcal{H}_e$ be the operator adding an e -ribbon with weight $h(-1)^h$, h the height. Then*

$$N_e = \mathcal{H}_e - M_{c_e}, \quad c_e = \frac{1}{2} \left[(e-1)p_e - \sum_{a+b=e, a,b \geq 1} p_a p_b \right],$$

and $\mathcal{H}_e = -\frac{d}{dt} R_e(t) \big|_{t=-1}$. Equivalently $N_e = M_{f'_e(-1)} - R'_e(-1)$, so $\text{ord } N_e = \text{ord } R'_e(-1)$: the order of N_e is the order of the first-order deformation of the ribbon operator away from the anchor $t = -1$, where $R_e(-1) = M_{p_e}$ has order 0.

Proof sketch. In the fermionic model $\alpha_{-a} = \sum_x \psi_{x+a} \psi_x^*$, and $\alpha_{-a} \alpha_{-b} = \alpha_{-(a+b)} - \Phi_{a,b}$ with $\Phi_{a,b} = \sum_{x,y} \psi_{x+a} \psi_{y+b} \psi_x^* \psi_y^*$. Put $\Psi_e := \frac{1}{2} \sum_{a+b=e} \Phi_{a,b}$, which is therefore the multiplication operator M_{-c_e} . Expanding Ψ_e over configurations, each term is a pair of bead hops $x_1 \rightarrow y_1, x_2 \rightarrow y_2$ with $(y_1 - x_1) + (y_2 - x_2) = e$, of two kinds. (i) *Genuine two-bead moves* ($\{y_1, y_2\} \cap \{x_1, x_2\} = \emptyset$): writing $x_1 < x_2, y_1 < y_2$, if $y_1 > x_2$ then both matchings $x_1 \rightarrow y_1, x_2 \rightarrow y_2$ and $x_1 \rightarrow y_2, x_2 \rightarrow y_1$ occur in the sum, and they differ by transposing $\psi_{y_1} \psi_{y_2}$, hence cancel; if $y_1 < x_2$ only the first occurs, the two hop intervals $(x_1, y_1], (x_2, y_2]$ are disjoint, and the resulting skew shape is a two-component generalized border strip of height $h_1 + h_2$, contributing $-(-1)^{h_1+h_2}$. (ii) *Chained pairs* ($y_1 = x_2$ or $y_2 = x_1$): the net effect is a single e -hop $x \rightarrow x + e$; for a fixed such hop the admissible chain points are exactly the h occupied sites strictly inside $(x, x + e)$, each contributing $(-1)^h$. Hence $\Psi_e = -N_e + \mathcal{H}_e$, which is the first display. The second follows from $R_e(t) = \sum_x \psi_{x+e} \psi_x^* \prod_{1 \leq i < e} t^{n_{x+i}}$ by differentiating at $t = -1$, and the last from differentiating $R_e = M_{f_e} - (1+t)E_e$ at $t = -1$ and using $M_{f_e(-1)} = M_{p_e} = R_e(-1)$. $\square$

Remark 15. Proposition 14 also explains the instrument test run at the start of the session: N_e must have order ≥ 1 , since order 0 would give $Q_{e,e'}(-1) = 0$ and contradict Theorem 2. Numerically $\text{ad}_{p_{a_1}} \cdots \text{ad}_{p_{a_r}}(N_e) \neq 0$ for all $r \leq 3$ and all distinct $a_i \in \{2, 3, 4, 5\}$, $e \in \{2, 3, 4\}$; the identity above shows where such order can come from, but the exact value of $\text{ord } N_e$ is not determined here and is not needed. See §8.

7 Verification

All computations are in `probes/2026-09-05-Q83/`. Two engines are used and they share no primitive: the *shape* engine (`nlib.py`, `ops.py`, built on `probes/2026-09-03-Q75/symfunc.py`, which enumerates skew shapes and computes components and heights from the Young diagram) and the *Maya* engine (`probes/2026-09-04-Q76/abacus.py` and `probes/2026-09-04-Q81/nested.py`, which works with β -sets and bead hops).

1. **Proposition 5** (`counts.py`): verified for all $\rho \in S_k$ and all $1 \leq r \leq k$, $2 \leq k \leq 8$ — 362,878 pairs (ρ, r) , 0 failures.
2. **Theorem 6** (`phi.py`): the closed form (3) was compared against the Maya engine's entry of $\mathcal{C}_k|\emptyset\rangle$ at $\mu = (E - j, 1^j)$ for *every* j in $0 \leq j \leq E - 1$, for 12 tuples with $k = 3, 4$ — 149 entries, 0 mismatches.
3. **Negative control.** Replacing $\Sigma_T + e_{\min} \leq j$ by $\Sigma_T + e_{\min} < j$ in (3) and rerunning check 2 on two tuples produces 6 mismatches out of 25 entries: the comparison is not blind.
4. **No distinctness.** Check 2 rerun on 9 tuples with repeated sizes among $e_1, \dots, e_{k-2}$ (e.g. $(2, 2, 3)$, $(3, 2, 2, 3)$, $(2, 2, 3, 3, 4)$) — 97 entries, 0 mismatches.
5. **Sharpness of the hypothesis.** For $(2, 3, 3)$, $(2, 3, 4, 4)$, $(5, 2, 2)$ the Maya engine returns $\mathcal{C}_k|\emptyset\rangle = 0$, confirming that $e_{k-1} = e_k$ is the exact degenerate case.
6. **Consistency with Theorem 2.** Evaluating (3) at $k = 3$, $j = f_2$ over all 60 distinct triples from $\{2, \dots, 6\}$ reproduces Theorem 2 exactly, including the vanishing when the largest size is outermost.
7. **Independent sharpness test.** The reduction operator $\text{ad}_{p_{e_1}} \cdots \text{ad}_{p_{e_{k-2}}}(Q_{e_{k-1}, e_k}(-1))$, built from the shape engine alone, is nonzero on $|\emptyset\rangle$ for all 12 tuples tested at $k = 3, 4$ including those with the largest size outermost, and the Maya engine returns $\text{gcd} = 1 + t$ literally for $(4, 2, 3)$, $(4, 3, 2)$, $(5, 2, 3)$, $(6, 2, 4)$, $(5, 3, 4)$.

8 Gaps and what remains

1. **The gcd versus its $(1 + t)$ -valuation.** Theorem 8 proves that $1 + t$ divides every entry and $(1 + t)^2$ does not divide all of them. The stronger literal statement “ $\text{gcd} = 1 + t$ ” additionally requires that no power of t (or other factor) divides every entry. For the tuples tested the gcd is literally $1 + t$, but this is **computed**, not proved; the same small gap is present in the corollary to Theorem 2. It would follow from exhibiting, for each $\vec{e}$, an entry with nonzero constant term in t ; Theorem 6 cannot supply one, since all its entries carry the factor $t^{j-1-|T|}$ and $j \geq e_{\min} \geq 2$.
2. **The order of N_e .** Proposition 14 identifies N_e but does not determine $\text{ord } N_e$. Numerically the iterated brackets $\text{ad}_{p_{a_1}} \cdots \text{ad}_{p_{a_r}}(N_e)$ are nonzero for $r \leq 3$; whether the order is infinite (as the coupling of a hop at x to the occupation at $x + i$ in the 2-quotient picture suggests, that being the diagonal which generates the unbounded moments $\sum_k k^m n_k$) is open. Theorem 8 makes this question independent of Q83.

3. **A second witness family.** The Maya engine shows entries $\pm t^a(t^2 - 1)$ at $\mu = (2, 2, 1^{E-4})$ for several tuples. These lie outside Lemma 3, whose chain lemma applies only to single-bead-hop targets: $(2, 2, 1^{E-4})$ is a two-bead-move target. A chain lemma for two-bead targets would give a second, independent witness family and is the natural next step.
4. **Sizes $e_i = 1$.** Everything above is stated for $e_i \geq 2$, matching the program. Nothing in the proofs uses $e_i \geq 2$ except the degree bound $e_{\min} \geq 2$ in Theorem 8, which only needs $e_{\min} \geq 0$; the case $e_i = 1$ was not tested.